

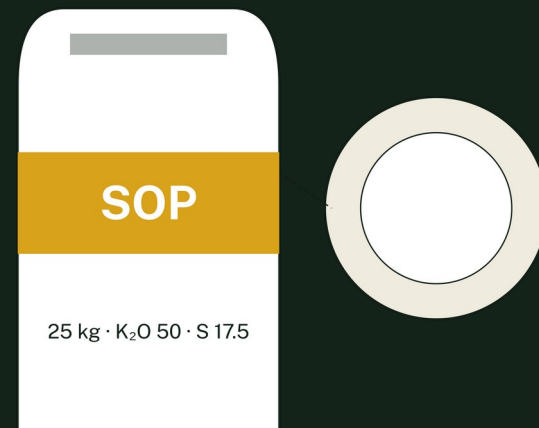


DEALER TRAINING · SEPTEMBER 2026

SOP

Sulphate of potash, K_2O 50%, S 17.5%, no chloride, 25 kg, for the soil and the water, never the drone.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

SOP is sulphate of potash: K₂O 50%, sulfur 17.5%, no chloride, in a 25 kg bag. It is the same potash as the commodity bag with sulfur in place of chloride. That matters on saline or sodic ground, where chloride adds to the salt the root already has to drink through, and on chloride-sensitive crops: potato, tobacco, most fruit and much of horticulture. The sulfur is sulfate, the form the root takes up as it is, and it counts in the season's sulfur. It is a basal granule: in the ground at land preparation, drilled at sowing or broadcast if there is no drill; where the grower fertigates, it dissolves and goes through the water. Never through a drone. SOP is a traded grade VAN supplies under its own name; the registration paper is VAN's to show, ask for it.

The 4 things it must contain

1

The bag: K₂O 50, S 17.5, no chloride, 25 kg. 12.5 kg of K₂O and 4.4 kg of sulfur in every bag.

2

The reason to buy it: what it does not contain. Chloride on saline ground and on chloride-sensitive crops is the case.

3

The sulfur: 17.5% as sulfate, counted in the programme, the nutrient that turns nitrogen into protein.

4

The method: soil at land preparation, drilled at sowing or broadcast, or through the water. Never the drone.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the bag says, and how it is registered.

25 kg

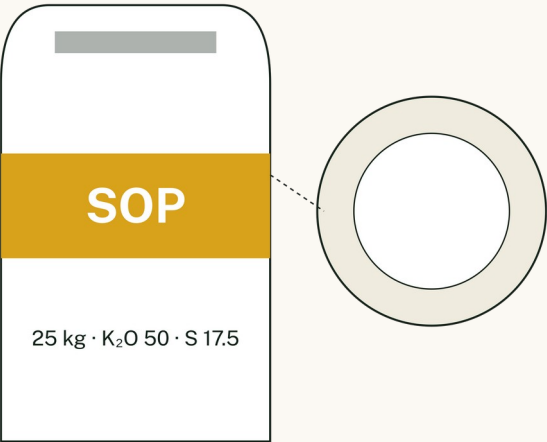
the bag. 12.5 kg of K₂O and 4.4 kg of sulfur in every bag, from the declared analysis.

K₂O 50%

potash, w/w, declared on the front.

S 17.5%

sulfur, w/w, as sulfate: the form the root drinks, ready as soon as it dissolves.



0 chloride

the defining property. Sulfate in place of chloride is the reason the bag costs more than the commodity potash.

Soil and fertigation

the 2 routes van.com.pk names. Not a drone or ultra-low-volume grade, and it will not become one.

Traded grade

SOP is a traded commodity to which no VAN manufacturing licence applies. The paper for the lot is VAN's to show, on request.

Registered as	Pakistan Standard	Manufacturing licence
SOP	Registration: ask VAN	Registration: ask VAN

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

“MOP is cheaper potash. Why would I pay more for SOP?”

It is the question at every counter where SOP sits beside the commodity bag, and the honest answer starts by agreeing: unit for unit, the commodity potash is cheaper potassium. The reason to buy SOP is what it does not contain, and what it carries instead. The answer is 3 ideas: chloride, sulfur, and where the bag goes.

The 3 answers on the next 3 slides, 1 idea each.

Same potash. No chloride. That is the reason.

Muriate of potash is potassium chloride. The crop needs a little chloride and gets it from rain and canal water; the rest is a passenger in the soil solution. On saline or sodic ground, and 9.0% of Punjab samples read saline, every extra salt makes it harder for the root to draw water. On chloride-sensitive crops, potato, tobacco, most fruit and much of horticulture, chloride itself costs quality. SOP is potassium sulphate: the same K_2O with sulfur in place of chloride, and a lower salt load per bag. Buy it for the chloride it does not have.

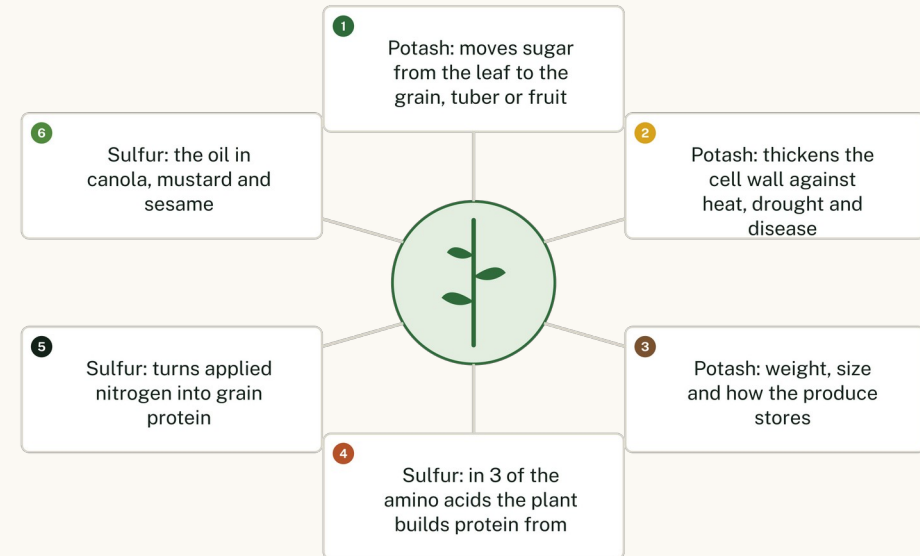


What is in each bag. MOP is a general product name; VAN prints no MOP price and no competitor brand.

Say it like this: “Unit for unit, MOP is cheaper potash. You buy SOP for what is not in it: no chloride, on salty ground and on crops that will not take it.”

The sulfur is a second nutrient, not a passenger.

Every bag carries 4.4 kg of sulfur as sulfate, the form the root takes up as it is. Sulfur is in 3 of the amino acids the plant builds protein from; without it the nitrogen a grower applied stays unfinished and leaves the field. It is also the oil in canola, mustard and sesame. Sulfate moves with the water, so it is counted in the season the bag goes on, and the programme's sulfur is reduced by what this bag brings. VAN's soil report does not test sulfur, so no national sulfur figure is claimed; van.com.pk says so.

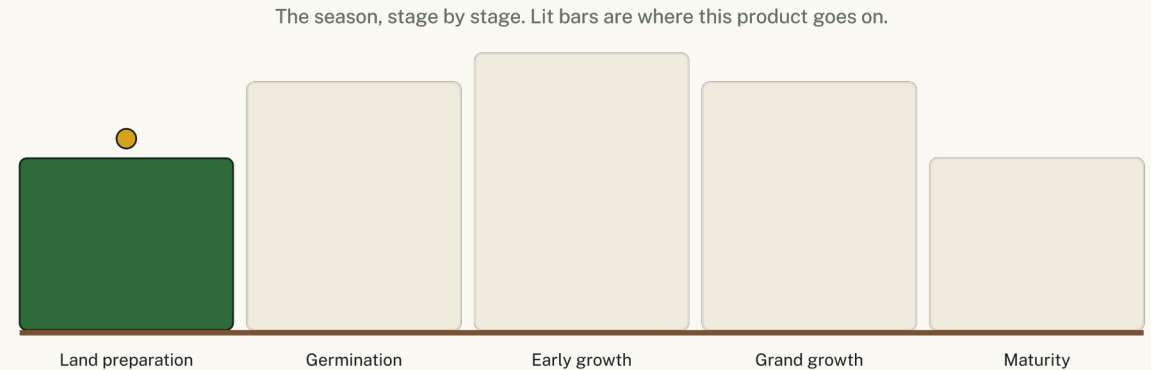


The 6 jobs are potash and sulfur from the 17-nutrients page on van.com.pk. A picture of the idea, not a measurement.

Say it like this: “There is sulfur in this bag, 4.4 kg, ready for the root. Count it in your programme instead of buying it again on top.”

In the ground at land preparation, or through the water. Never the drone.

SOP is a basal granule: it goes in the ground at land preparation, drilled beside and below the seed row first, side-dressed second, broadcast last. It dissolves well, so where the grower fertigates it goes through the drip or flood water at the potassium window. VAN's dose guidance of 23 July 2026 is ½ bag an acre on wheat and rice, 1 bag on sugarcane, citrus, potato and other crops; no crop programme prints an SOP row yet. It is a soil and fertigation product and not a drone grade, and it will not become one.



Land preparation is the stage for the basal bag; the rates are VAN's guidance until a programme row prints them.

Say it like this: “In the ground before the crop, or dissolved in the water. ½ bag on wheat and rice, 1 bag on most other crops, as VAN's guidance stands. Never in the drone.”

THE EVIDENCE

What is verified for SOP, and what is not.

SOP is a traded grade, and no VAN trial of it exists. What a dealer may state as fact is the bag, the registration position, the soil record and the published potash research, each as what it is.

Item	What it says	How a dealer may use it
The bag	SOP, sulphate of potash, K ₂ O 50% w/w, S 17.5% w/w, chloride-free, 25 kg. Soil and fertigation.	As fact.
Registration	SOP is a traded commodity to which no VAN licence applies, as the regulatory page on van.com.pk says. What VAN holds is the capacity to have it made under a vendor's own PSQCA licence.	Say "registration: ask VAN". Never quote a licence number for it.
The soil record (Punjab, 770,160 samples)	87.8% of samples below the working threshold for available potash; mean 132 ppm. 9.0% read saline; mean EC 2.07 dS/m. Sulfur: 0 samples tested.	As fact about the soil. Never as product performance.
National potash use	Potash is 1% of Pakistan's nutrient use (the Crop Force bag). About 1 kg of potash for every 83 kg of nitrogen (van.com.pk).	As fact.
Published research, not VAN trials	Sindh banana (Memon et al. 2010): correcting potash alone lifted yield from 51.2 to 60.8 t/ha. Multan cotton (Hussain, Ali and Gardezi 2021): soil potassium reduced boll shedding.	Spoken, as research on potash. Never as an SOP result.
Not on file	A trial of SOP; a programme row; a PS number or VAN licence; a batch certificate; a salt-index figure; a Pakistani sulfur deficiency figure.	Not said.

Sources: the bag and van.com.pk; the regulatory page on van.com.pk; VAN's statement of 20 Aug 2026 on making SOP under a vendor's licence; the Punjab soil testing programme (36 district workbooks, 770,160 samples); Memon et al. 2010; Hussain, Ali and Gardezi 2021.

The bag, in the grower’s own back-of-envelope.

Price from the VAN farmer price sheet, Aug 2026, delivered farmer price. K₂O and sulfur from the declared analysis. Rates are VAN’s dose guidance of 23 July 2026; no programme row prints them yet.

SOP, K ₂ O 50 · S 17.5	Weight	K ₂ O	Sulfur	Rupees
1 bag	25 kg	12.5 kg	4.4 kg	8,000
Wheat or rice, ½ bag an acre	12.5 kg	6.25 kg	2.2 kg	4,000
Sugarcane, citrus, potato, 1 bag an acre	25 kg	12.5 kg	4.4 kg	8,000

On wheat the grower spends 4,000 rupees an acre to put 6.25 kg of K₂O and 2.2 kg of sulfur in the ground, with no chloride. That is the counter arithmetic. What the crop does with it is not a number VAN prints.

SOP 25 kg PKR 8,000: the VAN farmer price sheet, Aug 2026. 12.5 kg = 25 × 0.50; 4.4 kg = 25 × 0.175. Rates: VAN’s dose guidance of 23 July 2026.

The potash account for an acre of wheat, and the sulfur that comes with it.

What 40 maunds of wheat (1.6 tonnes) takes off the field, from the IPNI removal tables the site’s balance block uses, beside ½ bag of SOP.

Wheat, 1 acre at 40 maunds	K ₂ O at 1.6 t	Sulfur at 1.6 t	
Grain taken off (K ₂ O 4.8, S 1.7 per tonne)	7.7 kg	2.7 kg	
Straw (bhusa) taken off too (K ₂ O 20, S 2.3 per tonne of grain)	32 kg	3.7 kg	
Grain and straw together	40 kg	6.4 kg	
SOP, ½ bag at land preparation, puts in	6.25 kg	2.2 kg	

On a field where the bhusa leaves too, the crop takes off about 40 kg of K₂O and 6.4 kg of sulfur. ½ bag of SOP puts back 6.25 kg and 2.2 kg. The rest is the soil’s reserve, and 87.8% of Punjab samples already read low in potash. A case for the basal bag in the grower’s own numbers, and not a yield claim.

IPNI nutrient removal tables (metric, 01/15): wheat grain K₂O 4.8 and S 1.7; straw K₂O 20 and S 2.3 per tonne of grain. 40 maunds × 40 kg = 1.6 t. Soil: the Punjab soil record, 87.8% below threshold. No Pakistani removal table exists on the public record, as van.com.pk says.

The 3 potash bags are 3 jobs, not 1 choice.

Bag prices side by side from the VAN farmer price sheet, Aug 2026. The dealer sells the stage and the job, never the kilogram.

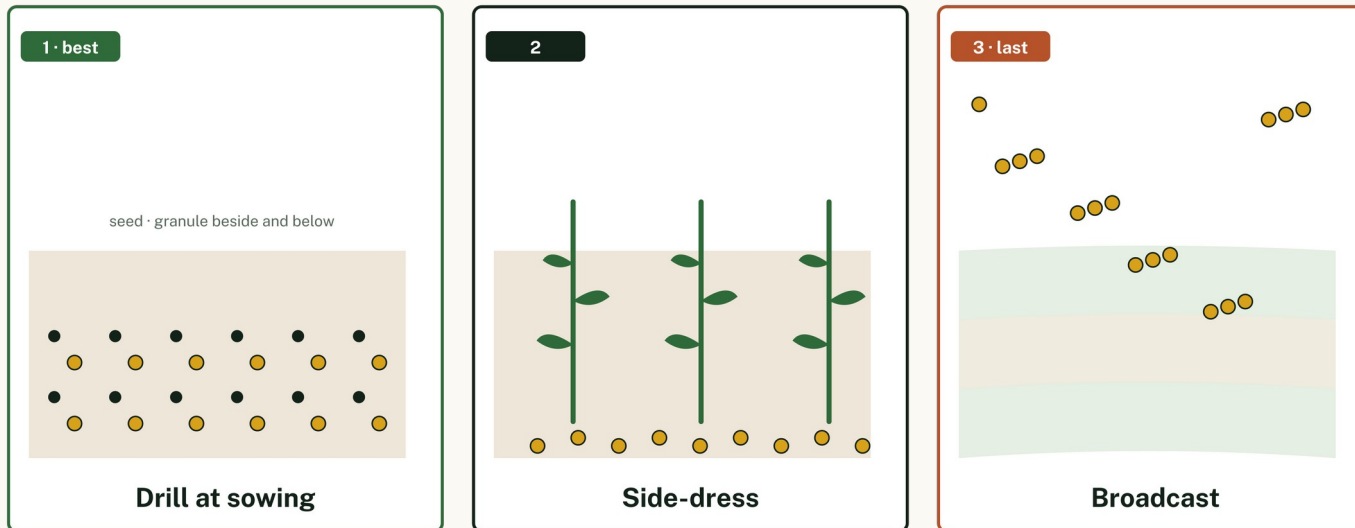
Product	Bag	K ₂ O in the bag	Stage and method	Rupees
SOP, K ₂ O 50 · S 17.5, no chloride	25 kg	12.5 kg	Land preparation, drilled or broadcast; or through the water	8,000
Fusion Potash, K ₂ O 50 with S, B, Mg	25 kg	12.5 kg	Land preparation, drilled or broadcast	11,500
Vital Potash, N 11 · K ₂ O 44	20 kg	8.8 kg	Grand growth and maturity, through the water	10,500

SOP and Fusion Potash are both basal bags: SOP where chloride or salt is the problem and sulfur is wanted; Fusion Potash where boron and magnesium are wanted on the granule. Vital Potash is the fill line. MOP is not on the VAN price sheet, so no price is printed for it.

Prices: the VAN farmer price sheet, Aug 2026, SOP 25 kg 8,000; Fusion Potash 25 kg 11,500; Vital Potash 20 kg 10,500. K₂O per bag from the declared analyses.

HOW IT GOES ON

Basal, in the ground at land preparation. Through the water where the grower fertigates. Never the drone.



1. Drill at sowing, best

The seed drill's fertilizer box places the granule about 5 cm beside and below the seed row, never touching the seed. The root meets it first; the whole bag is in the root zone.

2. Side-dress, second; broadcast, last

A band beside the row, covered with soil, at sowing or the first irrigation. Or spread before the last ploughing and mixed in: the fallback where there is no drill, and the last in the order. On the surface it feeds the surface.

Through the water, where the field is fertigated

SOP dissolves well: into the drum at the head of the watercourse or the drip tank, at the potassium window. Soil and fertigation are the 2 routes. It is not a drone or ultra-low-volume grade. Whatever else is flown, this is not.

Drill first, side-dress second, broadcast last.

The order for every basal granule VAN sells. The same bag gives a different result by where it lands. SOP also goes through the water where the grower fertigates.

Order	Method	What it means	Why
1 · best	Drill at sowing	The seed drill's fertilizer box places the granule about 5 cm beside and below the seed row, never touching the seed.	The root meets it first. Least contact with air, sun and free calcium at the surface. The whole bag is in the root zone.
2	Side-dress	A band beside the row, by hand or planter, at sowing or the first irrigation, covered with soil.	Still a band near the root, but later and shallower than a drill.
3 · last	Broadcast	Spread on the surface before the last ploughing or before an irrigation, and mixed in.	Most of the bag is away from the root, and most of it is exposed at the surface.

In VAN's own fields and partner fields, on wheat, maize and sugarcane, basal fertilizer drilled at sowing has given 7 to 9% better response than the same bag broadcast: the mean of replicated observations across regions, not a published trial.

For the soil application. Where it goes through the water, follow the fertigation rate.

OBJECTION 1 OF 7

“8,000 for 25 kg? MOP is cheaper and it is the same 50% potash.”

The dealer says, out loud

Yes. Unit for unit, MOP is cheaper potassium, and I will not tell you otherwise. You buy SOP for what is not in it. MOP is potassium chloride; on salty ground and on potato, fruit and vegetables the chloride costs you. SOP has none, and it carries 4.4 kg of sulfur a bag that you would otherwise buy separately. If your ground is sweet and your crop takes chloride, the cheaper bag may be the right bag. Tell me the field and the crop.

The fact behind it

- SOP 25 kg PKR 8,000 (VAN farmer price sheet, Aug 2026). MOP carries no price on the VAN sheet.
- 12.5 kg K₂O and 4.4 kg S per bag (the declared analysis); MOP is potassium chloride, K₂O 60, no sulfur.
- Chloride-sensitive crops: potato, tobacco, most fruit, much of horticulture (general agronomy).

Never say: “SOP gives more yield than MOP.” No trial on VAN’s record says so.

“Can I fly it in the drone with my spray?”

The dealer says, out loud

No. SOP is a soil and fertigation product. It goes in the ground at land preparation or dissolved through the irrigation water. It is not a drone or ultra-low-volume grade and VAN will not present it as one. If a product is meant to be flown, VAN names it into its drone family; this one is not in it.

The fact behind it

- SOP is a soil and fertigation product, not a drone grade, and it will not become one (the application systems page on van.com.pk).
- The product page names 2 routes: soil and fertigation (van.com.pk).
- A product meant to be flown is named into VAN's One Tank family; SOP is not in it (van.com.pk).

Never say: “It dissolves, so it will fly.”

“Which stage do I put it on?”

The dealer says, out loud

In the ground before the crop, at land preparation, drilled beside and below the seed row if you drill, broadcast and mixed in if you do not. If your field is on drip or you fertigate, dissolve it and run it with the water at the potassium window. VAN's guidance of 23 July is ½ bag on wheat and rice and 1 bag on sugarcane, citrus, potato and most other crops. No crop programme prints an SOP row yet, so the guidance is what I have.

The fact behind it

- SOP is a basal granule: land preparation, drilled at sowing or broadcast if no drill.
- Rates: VAN's dose guidance of 23 July 2026, wheat and rice ½ bag, sugarcane, citrus, potato and other crops 1 bag.
- No published crop programme on van.com.pk prints an SOP row.

Never say: A stage or rate the guidance does not name, or a foliar rate.

“Is it registered? Where is VAN’s licence for it?”

The dealer says, out loud

SOP is a traded grade, and van.com.pk says so plainly: it is a commodity to which no VAN manufacturing licence applies. What VAN holds is the capacity to have it made under a vendor’s own PSQCA licence. For the paper that covers the bag you are buying, ask VAN; I will not quote a licence number that is not on this product.

The fact behind it

- SOP is a traded commodity to which no VAN licence applies (the regulatory page on van.com.pk).
- VAN can have SOP made under a vendor’s own PSQCA licence (VAN, 20 Aug 2026).
- 20 of VAN’s 23 brands hold a live PSQCA manufacturing licence; SOP is 1 of the 3 with a stated reason (van.com.pk).

Never say: A PS number or a CM/L number for SOP.

OBJECTION 5 OF 7

“My land is salty. Will this fix it?”

The dealer says, out loud

It will not fix salinity, and VAN does not say it will. What it does is put potash and sulfur on without adding chloride to a soil solution that is already salty, so the root is not asked to drink through more salt to get its potash. On saline or sodic ground that is the reason the bag is chosen. The salinity itself is a drainage and water question, and the soil report is where it starts.

The fact behind it

- 9.0% of Punjab samples read saline; mean EC 2.07 dS/m (the Punjab soil record, 770,160 samples).
- Sulphate of potash carries a lower salt load per bag than potassium chloride (general soil chemistry; no salt-index figure is printed).
- No VAN potash claims a soil-test or EC change.

Never say: “It cures saline soil” or “it lowers your EC”.

“If there is sulfur in it, do I still need Green Sulfur?”

The dealer says, out loud

Count what this bag brings: 4.4 kg of sulfur a bag, as sulfate, ready for the root this season. Then read the programme. Where the programme puts Green Sulfur on for the season's sulfur, that row stands; where SOP is on the field, the sulfur it brings is part of the account, not on top of it. VAN does not say one replaces the other; the programme and the field decide.

The fact behind it

- 4.4 kg S per bag = 25×0.175 .
- Sulfate is the form the root takes up; elemental sulfur must be turned into sulfate by soil bacteria first (the 17 nutrients page on van.com.pk).
- Wheat at 40 maunds takes off about 6.4 kg S with the bhusa (IPNI removal tables).

Never say: “SOP replaces Green Sulfur” or “you never need sulfur again”.

“Is SOP organic? Can I use it on my organic plot?”

The dealer says, out loud

VAN does not print “organic” on this bag and I will not say it. SOP is a mineral potash, sulphate of potash. Whether a particular certifier accepts it on an organic plot is that certifier’s rule, not VAN’s claim. Ask the certifier, with the bag in hand.

The fact behind it

- The word “organic” is not on the bag or on the product page (van.com.pk).
- SOP is a mineral salt, potassium sulphate (general chemistry).
- VAN makes no organic claim for any potash.

Never say: “It is organic-approved.”

SOP is not yet a row in a published programme.

No published crop programme on van.com.pk prints an SOP row yet, so no rate per crop is published. VAN's dose guidance of 23 July 2026 is ½ bag an acre on wheat and rice and 1 bag an acre on sugarcane, citrus, potato and other crops, in the ground at land preparation or through the water where the field is fertigated. A dealer quotes it as VAN's guidance and sends the grower to VAN for a written plan.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1

A yield figure, or that SOP beats MOP. No trial on VAN's record.

2

That it fixes or cures saline soil, or lowers the soil's EC. It adds no chloride; it does not change the field.

3

A drone or ultra-low-volume use. Soil and fertigation only, and it will not become a drone grade.

4

A VAN licence or Pakistan Standard number. It is a traded grade; the paper is VAN's to show, on request.

5

"Organic." The word is not on the bag and VAN does not claim it.

6

That it replaces Green Sulfur or the fill line. The sulfur is counted in the programme; the programme decides.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a grower may ask for on an SOP bag

The declared analysis, K₂O 50% and S 17.5%, and the lot or batch mark on the bag. SOP is a traded grade: no VAN manufacturing licence applies to it, and no batch certificate for an SOP lot was on file for this deck. The dealer says “registration: ask VAN” and sends the batch mark to VAN for the paper that covers that lot.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

GUARD. The promise word for potash.

Urdu first pass, unreviewed

فصل کی ڈھال

The same potash, without chloride, with sulfur.

Urdu first pass, unreviewed

وہی پوٹاش، کلورائیڈ کے بغیر، سلفر کے ساتھ

In the soil at land preparation, or with the water. Never by drone.

Urdu first pass, unreviewed

زمین کی تیاری پر مٹی میں دیں یا پانی کے ساتھ، ڈرون سے کبھی نہیں

Buy it because of what is not in it.

Urdu first pass, unreviewed

اسے اس لیے خریدیں کہ اس میں کلورائیڈ نہیں

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

SOP: the same potash without the chloride, with sulfur. In the ground at land preparation or through the water. Never the drone.

